Q1 12% p.a. effective $= (1.12)^{1/2} - 1 = 5.830052\%$ per half year effective [1]

a) $i^{(2)} = 2 \times 5.830052\% = 11.6601\%$ [1]

b) $d = \frac{i}{1+i} = \frac{0.05830052}{1.05830052} = 5.5089\%$ [1]

c) present value $= 100 \times \frac{1}{d} = \frac{100}{0.055089} = £1,815.25$ [1]

Total [4]

Q2 a) · Value of first year's cashflows at $t=3$

$= 500(1+i)^3 + 500(1+i)^{2\frac{1}{2}}$ [1] @ $i \neq 3\%$ [½]

$= £1084.71$ [½]

· Value of second year's cashflows at $t=3$

$= 200\ddot{s}_{\overline{4}|} \times (1.03)$ [1] with $\ddot{s}_{\overline{4}|}$ calculated at

$i = (1.03)^{1/4} - 1$

$= 0.7417070177\%$ [½]

$= 200\left[\frac{(1.007417070177)^4 - 1}{0.007417070177}(1.007417070177)\right] \times 1.03$ [1]

$= 200 \times 4.07472289 \times 1.03$

$= £839.39$ [½]

· Value of third year's cashflows at $t=3$

$= 100 s_{\overline{12}|}$ [1] with $s_{\overline{12}|}$ calculated at

$i = (1.03)^{1/12} - 1$

$= 0.2466269777\%$ [½]

$= 100\left[\frac{(1.002466269777)^{12} - 1}{0.002466269777}\right]$ [1]

$= 100 \times 12.16411942$

$= £1,216.41$ [½]

→ Value of account at $t=3 = 1084.71 + 839.39 + 1216.41$

$= £3,140.51$ [1]

Q2

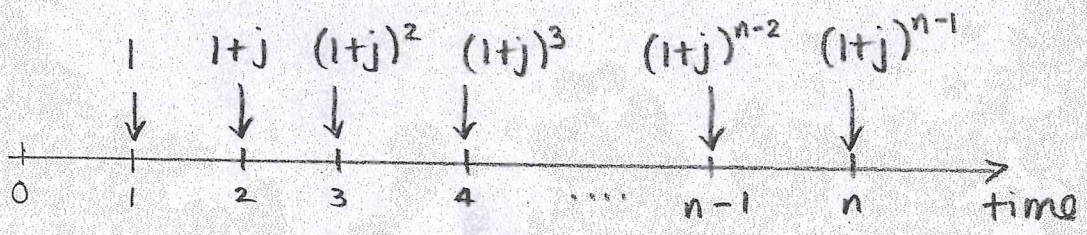
b) $p.v. = \dfrac{3,134.33}{(1.03)^3} = £2,868.36$ [1]

c) The answer would be <u>lower</u> if interest was simple. [1]

This is because no <u>further</u> interest is received on the <u>previous</u> interest received. [1] [Total: 12]

Q3

a)



$p.v. = 1.v + (1+j)v^2 + (1+j)^2 v^3 + \ldots + (1+j)^{n-1} v^n$ [2]

$= \dfrac{1}{(1+j)} \left[\dfrac{(1+j)}{(1+i)} + \dfrac{(1+j)^2}{(1+i)^2} + \ldots + \dfrac{(1+j)^n}{(1+i)^n} \right]$ [1]

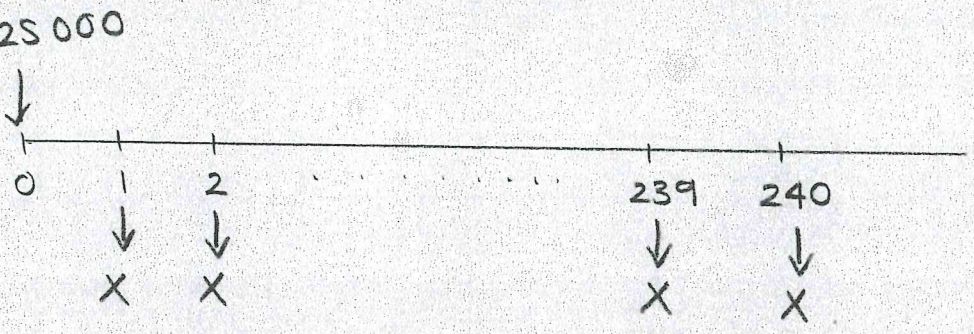
Let $(1+i') = \dfrac{(1+i)}{(1+j)}$ [1]

$\rightarrow p.v. = \dfrac{1}{(1+j)} \left[\dfrac{1}{(1+i')} + \dfrac{1}{(1+i')^2} + \ldots + \dfrac{1}{(1+i')^n} \right]$ [1]

$= \dfrac{1}{(1+j)} a_{\overline{n}|}^{i'}$ [1]

Q3

b)



Let X be the level monthly repayment

$$X = \frac{125{,}000}{a_{\overline{240}|}^{i}} \quad [1]$$

where $i = (1.06)^{1/12} - 1 = 0.486755056\%$ [1]

$$a_{\overline{240}|}^{i} = \frac{1 - \left(\frac{1}{1.00486755056}\right)^{240}}{0.00486755056} = 141.3843005 \quad [1]$$

$$\rightarrow X = \frac{125\,000}{141.3843005} = £884.12 \quad [1]$$

c) i) Capital outstanding after 120th repayment

= p.v. of future repayments

$= 884.12\, a_{\overline{120}|}^{i}$ [1]

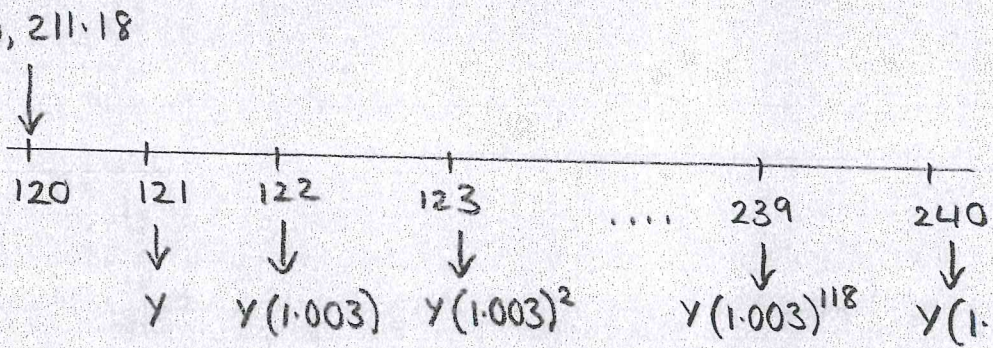
$$a_{\overline{120}|}^{i} = \frac{1 - \left(\frac{1}{1.00486755056}\right)^{120}}{0.00486755056} = 90.72431399 \quad [1]$$

$\rightarrow$ capital o/s $= 884.12 \times 90.72431399$

$= £80,211.18$ [½]

Q3

c) ii)



Let Y be the first repayment after terms change

→ p.v. of future repayments

$= Y \cdot \frac{1}{1.003} \cdot a_{\overline{120}|}^{i'}$ [1] (using result from part a))

where $i' = \frac{1+i}{1+j} - 1$

$= \frac{1.004867550 56}{1.003} - 1$

$= 0.1861964\%$ [1]

→ $a_{\overline{120}|}^{i'} = \frac{1 - \left(\frac{1}{1.001861964}\right)^{120}}{0.001861964} = 107.4497115$ [1]

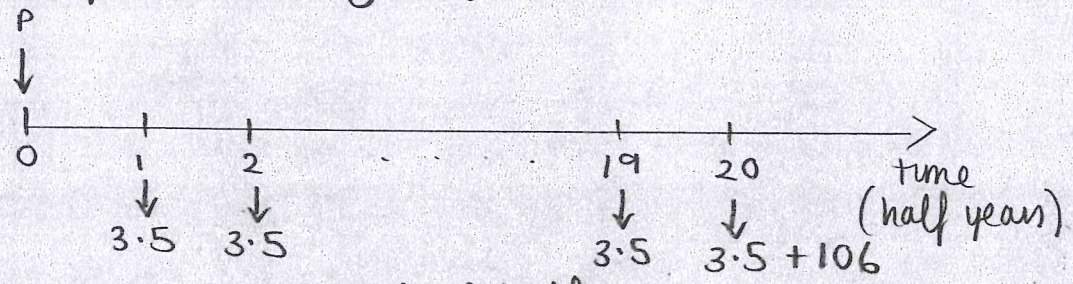
→ p.v. of future repayments

$= \frac{107.4497115}{1.003} Y = 80,211.18$ (by principle of equivalence) [1]

→ $Y = £748.74$ [½]

Q4

a) Cashflows (ignoring tax)



Working in time units of half years

$i^*(P) = (1.07635)^{1/2} - 1 = 0.037472891$ [½]

$\frac{g(1-t_1)}{R} = \frac{3.5(1-0.2)}{1.06} = 0.026415$ [1]

$< i^*$ so CGT is payable [½]

$\rightarrow P = 3.5(1-0.2)a_{\overline{20}|} + [106 - 0.3(106-P)]v^{20}$

@ $i^* = 0.037472891$ [2]

$a_{\overline{20}|} = 13.8996$ [½]

$\rightarrow P = 2.8 \times 13.8996 + \frac{74.2}{(1.037472891)^{20}} + \frac{0.3}{(1.037472891)^{20}} P$ [½]

$\rightarrow 0.856257P = 74.47126$ [½]

$\rightarrow P = £86.97$ [½]

b) Net real redemption yield

$= \frac{(1 + \text{Net monetary redemption yield})}{(1 + \text{inflation rate})} - 1$

$= \frac{1.07635}{1.03} - 1$

$= 4.5\%$ p.a. effective [1]

Total: [7]

Q5

a) The DPP is the time when the value of the investor's account changes from negative to positive

OR

The DPP is the time when the project starts to make a profit for the investor. [1]

b) Work in £ millions.

p.v. of cash in = 1

We require the lowest integer value of n such that

p.v. of cash in $\leq$ p.v. of cash out

$$1 \leq 0.1 \cdot \frac{1}{(1.07)^{10}} (I\ddot{a})^{7\%}_{\overline{n}|} \quad [2]$$

$$(I\ddot{a})_{\overline{n}|} = \frac{\ddot{a}^{7\%}_{\overline{n}|} - \frac{n}{(1.07)^n}}{0.07}(1.07) \quad [1]$$

For $n = 6$ $\rightarrow$ $\ddot{a}^{7\%}_{\overline{6}|} = 5.100197436$ [½]

$\rightarrow$ $(I\ddot{a})_{\overline{6}|} = 16.84705971$ [1]

$\rightarrow$ p.v. of cash out $= \frac{0.1}{(1.07)^{10}} \times 16.84705971 = 0.856 < 1$ [½]

For $n = 7$ $\rightarrow$ $\ddot{a}^{7\%}_{\overline{7}|} = 5.76653966$ [½]

$\rightarrow$ $(I\ddot{a})_{\overline{7}|} = 21.51145528$ [1]

$\rightarrow$ p.v. of cash out $= \frac{0.1}{(1.07)^{10}} \times 21.51145528 = 1.0935 > 1$ [½]

c) DPP takes no account of the overall level of profit achieved by the project [1]
A more suitable measure is Net Present Value [1]

Total: [10]

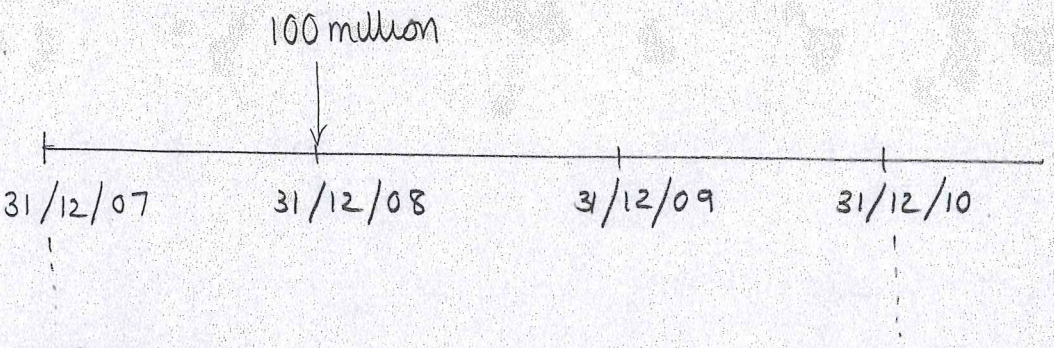
Q6

a)

Disadvantage of MWRR over TWRR:

→ MWRR is distorted by the amount and timing of cashflows, which may not be in manager's control [1]

b)



(in)

Let MWRR = i

Equation of value formulated at 31/12/10:

$f(i) = 157(1+i)^3 + 100(1+i)^2 - 301 = 0$ [1]

At $i = 6\%$ → $f(i) = -1.650488$ [1]

$i = 7\%$ → $f(i) = 5.821751$ [1]

$$\rightarrow i^* \approx 0.06 + (0.07-0.06)\left[\frac{0 + 1.650488}{5.821751 + 1.650488}\right]$$

$= 0.0622$ [1]

Q6 c)

Let F be the fund value on 31/12/08
Let i_1 be the yield on the fund between 31/12/07 & 31/12/08
Let i_2 be the yield on the fund between 31/12/08 & 31/12/10

Then : $157(1+i_1) = F$ [1]

$(F+100)(1+i_2)^2 = 301$ [1]

We know : $(1+i_1)(1+i_2)^2 = (1.0622)^3$

$\rightarrow \quad \frac{F}{157} \times \frac{301}{(F+100)} = (1.0622)^3$ [2]

$\rightarrow \quad \frac{301F}{157F + 15700} = (1.0622)^3$

$\rightarrow \quad 112.8438F = 18815.62044$

$\rightarrow \quad F = 166.74$ [1]

So the fund value on 31/12/08 = £166.74 million.